

huge benefit. Sharma adds, "VoIP networks are very scalable, meaning they can be expanded with relative ease given they use standard interfaces. It is also much easier to introduce new services over IP networks because of the way the protocols are structured. Also from a maintenance and operational perspective, IP networks are much easier to handle." Moreover, in order to scale a traditional circuit-switched network, additional lines would have to be added, but VoIP rules out the need for any such investment.

VoIP is also a very reliable technology. In a PSTN network, if a line is down, it is completely cut off and calls cannot be made from or to telephones in that line. However, in the case of VoIP, multiple routes can be established to reach a node, and hence downtime is negligible or nil.

VoIP is clearly the future of telephony as it combines a bouquet of services ranging from video conferencing to call-forwarding at a very low cost. With the spread of broadband as well as user awareness, VoIP is slowly picking up at the user level in India, but it is mainly the enterprises that have gone gaga over VoIP for both long-distance calling as well as internal communications.

Not a bed of roses

VoIP seems irresistible, but has still not reached traction on a mass scale in India, or even in many developed countries for that matter. One of the primary reasons is that VoIP is based on networks that are fundamentally different from the existing PSTN ones. Since the existing circuit-switched networks have still not served their life, no operator is prepared to sink those and invest in new IP-based networks.

Even otherwise, VoIP is still not as impeccable as it is portrayed to be.

"A circuit-switched network is deterministic. It has set a carrier-grade standard, especially in developed countries, which is hard to match by VoIP. A packet-switched network on the other hand is statistical, but allows tremendous flexibility for

VoIP vs PSTN

When somebody refers to 'traditional' telecom as against VoIP, he is likely to be referring to dear old PSTN!

Our nation's telephone network, and that of most others for that matter, works over what is known as the public-switched telephone network or PSTN. Some also call it the plain-old telephone service (POTS).

There is nothing so plain about it though. It is a worldwide network of telephone lines, fibre-optic cables, microwave transmission links, cellular networks, communications satellites and undersea telephone cables. This network is interconnected by switching centres that enable telephones across the world to connect to each other using a uniform numbering system. The whole setup is based on circuit-switched networking—that is, a circuit or channel is established between the various nodes and terminals before the users communicate, as if the nodes were physically connected with an electrical circuit.

The PSTN is a reasonably tech-savvy, digital system that includes mobile as well as fixed telephones. It is governed by various standards defined by the International Telecommunication Union's network standardisation sector (ITU-T).

However, with the growing popularity of Internet Protocol (IP) based networks, especially the Internet, varied methods, tools, protocols and standards have emerged to enable the transmission of voice and multimedia content over these networks. Essentially, voice over IP or VoIP is a bundle of technologies that enables IP-based network to be used as communication networks too. While PSTN is circuit-switched, VoIP is packet-switched. It sends voice or multimedia as data packets on an IP network.

handling converged multimedia applications, VoIP being one of them. As long as the legacy voice network is somewhat cost-effective and voice is the primary application, traditional circuit-switched voice will continue to dominate," opines Dr Suresh Borkar, senior principal investigator, Roberson and Associates, and faculty member, Illinois Institute of Technology, Chicago.

Dr Borkar adds that VoIP is not as competitive as it seems with regard to feature richness, reliability or quality. It, of course, potentially utilises the resources better since it sends packets as required, instead of reserved resources as in a circuit-switched voice connection. But the fundamental premise for the IP protocol is only its 'best effort' and so real-time applications like voice ultimately need to use the seemingly non-robust protocols like the user datagram protocol/real-time transport protocol (UDP/RTP) at higher levels. Plus, circuit infrastructure has also consistently been meeting the 'five nine' (99999) standards, whereas once again we must remember that the underlying basis for IP has been only 'best effort'.

So there is really no strong enough reason for service providers to ditch

their existing circuit-switched networks in favour of VoIP.

A mobile dimension to VoIP

Another recent addition to VoIP technology is mobile VoIP or mVoIP.

The name is quite self-explanatory. Mobile VoIP is a term used to describe the delivery of VoIP services via a mobile handset that is connected to Wi-Fi, WiMAX or 3G cellular networks. The delivery of services will be handled by a third-party service provider.

Although still nascent, according to a recent Frost & Sullivan report (*Impact of Mobile VoIP on Next Generation Cellular Networks*), mobile VoIP is poised to become the principal transport for various access technologies. It might even end up threatening traditional telecom.

At the end of 2008, approximately \$605.8 million of mobile VoIP revenues were generated in North America, Europe, Asia Pacific and Latin America. This, according to the report, is expected to grow to \$29.57 billion by 2015.

However, the study points out that many cellular operators have prohibited the use of mobile VoIP over their cellular networks, with some imposing a surcharge to avoid cannibalisation of